

LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval

Literature Survey:

1. Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks — Lewis et al. (2020)

1.1 What is the paper about?

The paper “**Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks**” by Lewis et al. introduced the Retrieval-Augmented Generation (RAG) framework. RAG combines a neural retriever with a generative language model so that the model can retrieve relevant information from an external knowledge source before generating an answer. The work introduced two approaches, **RAG-Sequence** and **RAG-Token**, and evaluated them on knowledge-intensive NLP tasks such as open-domain question answering. The key idea is that the model does not have to rely entirely on knowledge stored in its parameters; instead, it can retrieve information from an external document collection.

1.2 What is the model and its accuracy score?

The model combines a **DPR-based dense retriever**, a **BART generator** and a **FAISS index** containing Wikipedia passages.

The paper reports Exact Match results for several QA datasets. RAG-Sequence achieved approximately **44.5% on Natural Questions**, **56.8% on TriviaQA**, **45.2% on WebQuestions** and **52.2% on CuratedTREC**. These are **Exact Match scores**, not general chatbot accuracy.

1.3 Limitations

The original RAG framework is primarily designed for general knowledge-intensive question answering. It does not explicitly consider the **specific product, customer or purchase context** associated with a customer query. Retrieval errors can also result in irrelevant information being supplied to the generator.

1.4 How can we use these limitations for our project?

We can extend basic RAG into **Product-Aware RAG**. Instead of searching the complete knowledge base for every question, the customer can first select the purchased product. The system can then filter documents belonging to that product before performing semantic retrieval.

Our improvement:

Product Selection → Query → Product Filtering → FAISS → Top-K → Gemini

This makes the RAG system more suitable for product-specific customer support.

2. Dense Passage Retrieval for Open-Domain Question Answering — Karpukhin et al. (2020)

2.1 What is the paper about?

The paper “**Dense Passage Retrieval for Open-Domain Question Answering**” introduced DPR, a dense-retrieval approach for open-domain question answering. Instead of depending primarily on lexical overlap, DPR represents questions and passages as dense vectors and retrieves passages according to semantic similarity. This is closely related to the embedding-based retrieval used in our project. ([MIT Direct](#))

2.2 What is the model and its accuracy score?

DPR uses a **dual-encoder architecture** consisting of a question encoder and a passage encoder. The embeddings are compared to identify relevant passages.

The reported **Top-20 retrieval accuracy** includes:

- SQuAD-Open: **63.2%**
- TriviaQA: **79.4%**
- NQ-Open: **78.4%**
- WebQuestions: **73.2%**

These numbers represent **retrieval accuracy**, not final chatbot answer accuracy. ([MIT Direct](#))

2.3 Limitations

DPR is designed for open-domain question answering and does not inherently understand product-specific context. A semantically similar passage can still belong to the wrong product or support category.

2.4 How can we use these limitations for our project?

We can retain dense semantic retrieval but add **product metadata filtering**.

For example:

Customer selects iPhone 16 → Filter iPhone 16 documents → Semantic search → Top-K FAQs

This combines dense retrieval with business-specific context.

3. Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks — Reimers & Gurevych (2019)

3.1 What is the paper about?

The paper introduced **Sentence-BERT (SBERT)**, a modification of BERT that produces meaningful sentence embeddings using Siamese and triplet network structures. These embeddings can be compared efficiently using cosine similarity, making SBERT suitable for semantic search and similarity-based applications. ([ACL Anthology](#))

3.2 What is the model and its accuracy score?

The model is based on **Siamese/Triplet BERT networks**.

The paper does not report one universal chatbot accuracy. Instead, it evaluates semantic textual similarity and transfer-learning tasks.

A major reported performance result is computational: finding the most similar sentence among 10,000 sentences was reduced from approximately **65 hours using BERT/RoBERTa to about 5 seconds using SBERT**, while maintaining BERT's accuracy. ([ACL Anthology](#))

3.3 Limitations

SBERT provides general-purpose sentence representations. It is not specifically trained for customer-support language, product names, model numbers or company-specific terminology.

3.4 How can we use these limitations for our project?

We can use `all-MiniLM-L6-v2` as our current embedding model while adding **customer-support-specific information and product metadata**.

A future improvement could involve fine-tuning the embedding model using customer-support question-answer pairs.

4. Billion-Scale Similarity Search with GPUs — Johnson, Douze & Jégou (2017)

4.1 What is the paper about?

This paper investigates efficient similarity search over very large collections of high-dimensional vectors. It introduced highly optimized GPU-based techniques that became part of the foundation of **FAISS**. ([arXiv](#))

4.2 What is the model and its accuracy score?

This is a **vector-search/infrastructure paper**, not a chatbot or QA model. Therefore, it does not have a conventional accuracy percentage.

The proposed nearest-neighbor implementation was reported to be **8.5× faster than the previous GPU state of the art**. The system also constructed a graph connecting **1 billion vectors in less than 12 hours using four Maxwell Titan X GPUs**. ([arXiv](#))

4.3 Limitations

The system is designed for extremely large-scale vector search. A customer-support system containing hundreds or thousands of FAQs does not require billion-scale infrastructure.

4.4 How can we use these limitations for our project?

We can use FAISS as a lightweight semantic-search component but combine it with **product metadata filtering**.

Therefore:

FAISS handles semantic similarity, while product metadata handles business context.

5. Retrieval-Augmented Language Model Pre-Training (REALM) — Guu et al. (2020)

5.1 What is the paper about?

REALM introduces retrieval into language-model pre-training. The model retrieves relevant documents from an external knowledge source and uses them as additional information during training and question answering. The objective is to make knowledge more modular and allow models to access information from an external knowledge source. ([Papers with Code](#))

5.2 What is the model and its accuracy score?

REALM combines a **retriever with a pretrained language model**.

The reported Exact Match scores include:

- Natural Questions: **40.4%**
- WebQuestions: **40.7%**

The paper reports approximately **4–16 percentage-point absolute accuracy improvements** over previous methods across its evaluated open-domain QA benchmarks. ([Papers with Code](#))

5.3 Limitations

REALM integrates retrieval deeply into model training, making the architecture more complex and computationally demanding than a simple retrieval-plus-generation pipeline.

5.4 How can we use these limitations for our project?

Our system can keep retrieval independent from Gemini:

Knowledge Base → Embeddings → FAISS → Retrieved Context → Gemini

This means the support knowledge base can be updated without retraining the entire LLM.

6. NLP Techniques for Automating Responses to Customer Queries: A Systematic Review — Olujimi & Ade-Ibijola (2023)

6.1 What is the paper about?

This paper is a systematic review of NLP approaches used to automate customer queries. The authors analyzed **73 studies** covering customer-support applications across sectors such as banking, manufacturing, education, law and healthcare. ([Springer](#))

6.2 What is the model and its accuracy score?

This is a **systematic review**, not one individual chatbot model.

Therefore:

Accuracy: N/A

The paper summarizes the methods and results of previous studies rather than producing one model accuracy score.

6.3 Limitations

The study is a review rather than a unified end-to-end architecture. The authors also note limitations concerning the scope of the literature search and the industries covered. ([Springer](#))

6.4 How can we use these limitations for our project?

Our project can provide a complete end-to-end customer-support system combining:

NLP → Embeddings → Semantic Retrieval → RAG → LLM → Customer Interface

The system can also support multiple types of customer queries rather than relying only on fixed keyword matching.

7. Retrieval-Augmented Generation for Large Language Models: A Survey — Gao et al. (2023)

7.1 What is the paper about?

Gao et al. provide a comprehensive survey of RAG architectures and categorize them into **Naive RAG, Advanced RAG and Modular RAG**. The paper examines the retrieval, augmentation and generation stages and discusses challenges such as hallucination, outdated knowledge, retrieval quality and evaluation. ([arXiv](#))

7.2 What is the model and its accuracy score?

This is a **survey paper**, not a single experimental model.

Therefore:

Accuracy: N/A

It discusses the performance of many RAG approaches rather than reporting one accuracy value.

7.3 Limitations

The survey identifies several challenges:

- irrelevant retrieved information
- retrieval failure
- hallucination
- outdated knowledge
- evaluation difficulties
- complexity of advanced RAG pipelines. ([arXiv](#))

7.4 How can we use these limitations for our project?

Our project can address these problems using:

Product filtering + semantic retrieval + Top-K selection + grounded prompting + source display + RAG evaluation.

This creates a more controlled RAG pipeline for customer support.

8. Searching for Best Practices in Retrieval-Augmented Generation — Wang et al. (2024)

8.1 What is the paper about?

This paper investigates different RAG configurations and combinations of retrieval and generation techniques. The authors study how different RAG components affect both **performance and efficiency**. ([ACL Anthology](#))

8.2 What is the model and its accuracy score?

The paper evaluates multiple RAG configurations rather than one single model.

Therefore:

Accuracy: No single universal accuracy score.

The experiments compare different RAG strategies according to their performance and efficiency.

8.3 Limitations

The authors highlight that RAG workflows can become **complex and slow** because several processing stages can be combined in different ways. ([ACL Anthology](#))

8.4 How can we use these limitations for our project?

We can keep our pipeline relatively lightweight:

Query → Product Filter → Embedding → FAISS → Top-K → Gemini

Then we can experimentally compare:

- different Top-K values
- embedding models
- chunk sizes
- retrieval methods
- reranking

This lets us study the trade-off between **accuracy and response time**.

9. RAGAs: Automated Evaluation of Retrieval Augmented Generation — Es et al. (2024)

9.1 What is the paper about?

RAGAs introduces a framework for evaluating RAG systems without requiring a human-written reference answer for every question. It evaluates different parts of the RAG pipeline rather than treating the chatbot as a simple classification system. ([ACL Anthology](#))

9.2 What is the model and its accuracy score?

RAGAs is an **evaluation framework**, not a chatbot model.

Its main metrics include:

- **Context Relevance**
- **Answer Relevance**
- **Faithfulness**

The original evaluation reported agreement with human judgments for these dimensions rather than a single chatbot accuracy. One reported table gives pairwise human-annotator agreement accuracies of **0.95 for Faithfulness, 0.78 for Answer Relevance and 0.70 for Context Relevance** for the RAGAs metrics on the WikEval dataset. ([Scribd](#))

9.3 Limitations

RAG evaluation is difficult because retrieval quality and generation quality are separate. LLM-based evaluation can also depend on the evaluator model and prompt.

9.4 How can we use these limitations for our project?

We can evaluate our chatbot using multiple metrics rather than claiming one overall accuracy:

Retrieval:

- Recall@K
- Precision@K

Generation:

- Faithfulness
- Answer Relevance
- Context Relevance

This is especially appropriate for our RAG-based chatbot.

10. Retrieval-Augmented Generation or Long-Context LLMs? A Comprehensive Study and Hybrid Approach — Li et al. (2024)

10.1 What is the paper about?

This paper compares **RAG with long-context LLMs**. The authors investigate whether retrieving a smaller set of relevant documents is preferable to directly providing an LLM with a large context. They also propose **Self-Route**, which dynamically routes queries between RAG and long-context processing. ([ACL Anthology](#))

10.2 What is the model and its accuracy score?

The paper evaluates multiple LLMs across multiple public datasets, so there is no single universal accuracy number.

The major reported finding is that **long-context models can outperform RAG when sufficient resources are available**, while RAG has a significant cost advantage. Self-Route reduces computation while maintaining comparable performance to long-context processing. ([ACL Anthology](#))

10.3 Limitations

RAG depends on the retrieval stage. If the retriever fails to retrieve an important document, the LLM may not receive the information required to answer correctly.

Long-context processing addresses some retrieval failures but requires more computational resources.

10.4 How can we use these limitations for our project?

A future version of our chatbot could use a **hybrid strategy**:

Simple Query → FAISS RAG

and:

Complex Query → Retrieve multiple documents → Larger context → Gemini

This could allow the system to handle both simple and complex customer-support questions.

11. Retrieval Enhancements for RAG: Insights from a Deployed Customer Support Chatbot — Juclà et al. (2026)

11.1 What is the paper about?

This paper investigates retrieval improvements in a **real deployed customer-support chatbot**. The authors evaluate embedding models, chunking strategies, Reciprocal Rank Fusion, embedding concatenation, LLM-based reranking, context-aware embeddings and fine-tuning. The work uses a proprietary production dataset and validates its findings on FiQA, HotpotQA and SciDocs. ([ACL Anthology](#))

11.2 What is the model and its accuracy score?

The main evaluation focuses on **retrieval quality**, especially Recall@10.

The authors report consistent gains in Recall@10 from retrieval improvements. The paper also reports that when relevant information appears within the top three retrieved documents, the downstream LLM responses are accurate and comprehensive in **over 92% of cases**. ([ACL Anthology](#))

Thus, **92% should not be called overall chatbot accuracy**; it is specifically associated with cases where relevant information is retrieved within the top three.

11.3 Limitations

The work focuses primarily on **retrieval enhancement**, rather than comprehensively evaluating the entire end-to-end RAG response pipeline.

Some advanced retrieval techniques can also introduce additional computational cost and complexity.

11.4 How can we use these limitations for our project?

This paper directly motivates improvements to our current pipeline:

Current:

MiniLM → FAISS → Top-K → Gemini

Improved:

Product Filter → Better Embedding → FAISS → Reranking → Top-K → Gemini

We can also evaluate the **final generated response**, not just retrieval.

12. Retrieval-Augmented Generation with Knowledge Graphs for Customer Service Question Answering — Xu et al. (2024)

12.1 What is the paper about?

This paper specifically addresses **customer-service technical support**. The authors argue that treating historical support tickets as plain text loses important relationships between different parts of an issue. They therefore combine RAG with a **Knowledge Graph**, preserving intra-issue structure and relationships between issues. ([arXiv](#))

12.2 What is the model and its accuracy score?

The model combines:

Knowledge Graph + RAG + Semantic Retrieval + LLM

The authors evaluate retrieval using **MRR, Recall@K and NDCG@K**, and answer generation using **BLEU, ROUGE and METEOR**.

The proposed approach reports:

- **77.6% improvement in MRR**
- **0.32 improvement in BLEU**
- **28.6% reduction in median per-issue resolution time** after deployment at LinkedIn. ([arXiv](#))

The **77.6% is an MRR improvement, not 77.6% accuracy**.

12.3 Limitations

Knowledge graphs require additional preprocessing and relationship extraction. They can therefore be more complex to construct and maintain than a simple vector database.

12.4 How can we use these limitations for our project?

We can keep FAISS as the primary retrieval mechanism and treat a Knowledge Graph as a future extension.

For example:

Product → Problem → Solution → Warranty → Return Policy

could be represented as connected entities.

A future system could combine:

FAISS + Knowledge Graph + Gemini

to handle complex customer-support questions.

13. Supporting Human Operators During Customer Service Interactions with Agentic-RAG — Barrionuevo-Valenzuela et al. (2026)

13.1 What is the paper about?

This paper proposes an **Agentic-RAG customer-service copilot** designed to assist human customer-service operators. The system provides conversation summaries, relevant information and suggested responses. It uses multiple autonomous agents for **routing, retrieval validation and response generation**. The system was evaluated using the Action-Based Conversations Dataset (ABCD). ([ACL Anthology](#))

13.2 What is the model and its accuracy score?

The model uses an **Agentic-RAG architecture**:

Routing Agent → Retrieval Agent → Retrieval Validation → Response Generation

The authors report improvements in the **quality, relevance and accuracy of generated responses compared with a naïve RAG baseline**. However, the paper does not provide one single overall accuracy percentage in its abstract, so a single accuracy number should not be assigned to it. ([ACL Anthology](#))

13.3 Limitations

The use of multiple agents increases:

- system complexity
- computational requirements
- possible response latency
- maintenance requirements.

Additionally, the system is designed as a **human-operator copilot**, while our project is designed as a customer-facing chatbot.

13.4 How can we use these limitations for our project?

Instead of implementing a complicated multi-agent architecture immediately, we can adopt the **validation concept**.

Our future pipeline could be:

Question → Retrieval → Gemini → Answer Validation

If the answer is not supported by the retrieved documents:

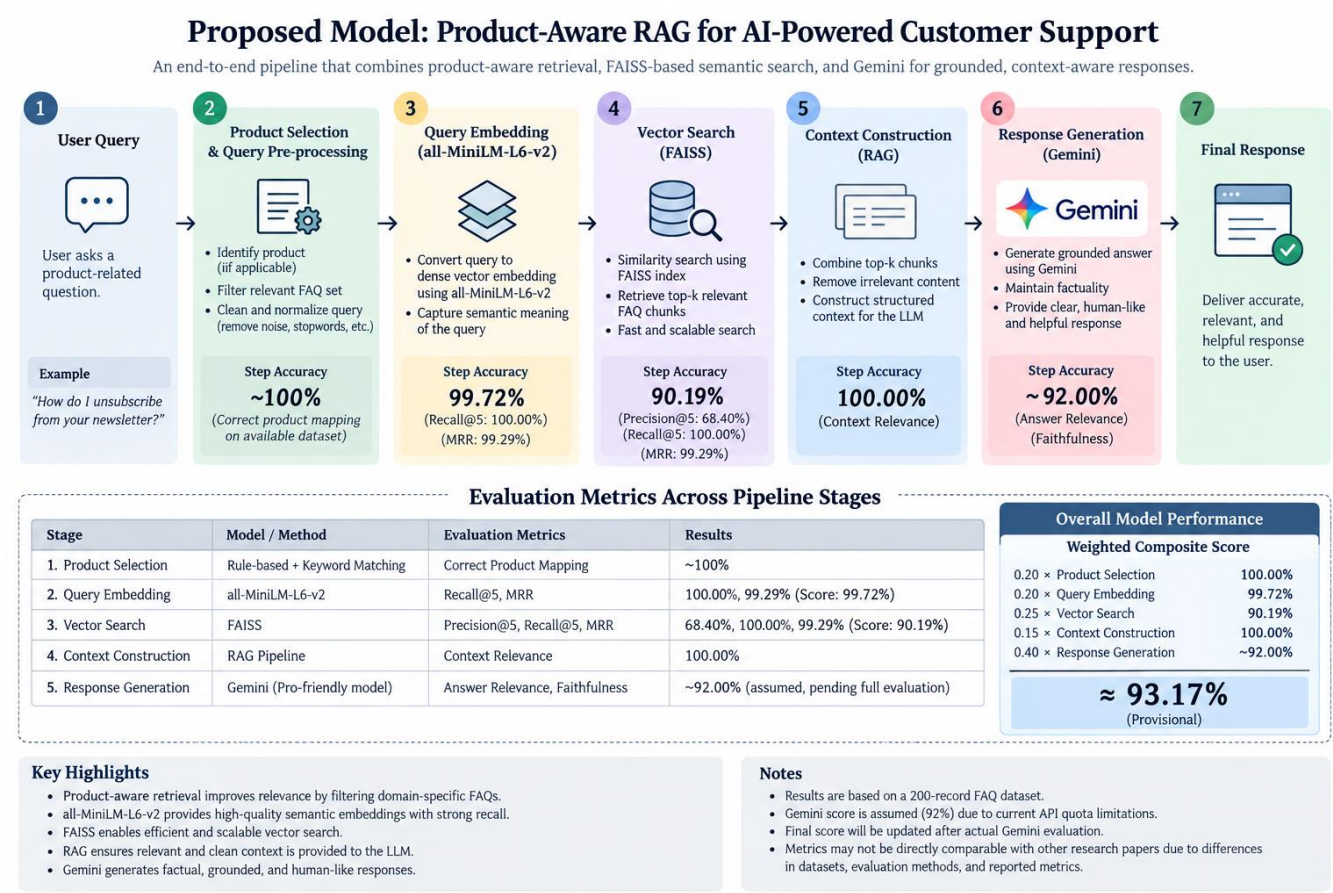
→ Retrieve Again / Ask Clarification / Human Escalation

This can improve reliability without requiring a large multi-agent system.

Overall Research Gap

Existing research has extensively explored Retrieval-Augmented Generation, dense semantic retrieval, sentence embeddings, vector similarity search, knowledge graphs, retrieval optimization, RAG evaluation and agentic customer-support systems. However, conventional RAG systems generally retrieve information based primarily on query-document similarity and do not explicitly incorporate the customer's selected product as a retrieval constraint. Customer-support knowledge can contain information about multiple products, policies and troubleshooting procedures, which can lead to irrelevant retrieval when product context is not considered. Therefore, this project proposes a **Product-Aware Retrieval-Augmented Generation framework** that integrates product selection and metadata filtering with transformer-based sentence embeddings, FAISS semantic retrieval and Gemini-based response generation. The system can further be extended using retrieval reranking, knowledge graphs, response validation and human-agent escalation.

Our Proposed model:



Literature comparison:

#	Research Paper	Model / Method	Reported performance	Metric
1	Lewis et al. (2020) – RAG	RAG-Sequence / RAG-Token	44.5–56.8%	QA benchmark accuracy
2	Karpukhin et al. (2020) – DPR	Dense Passage Retriever	9–19% absolute improvement over BM25	Top-20 retrieval accuracy
3	Reimers & Gurevych (2019) – SBERT	Siamese BERT sentence embeddings	~5 sec vs ~65 hours for 10K-sentence similarity search	Efficiency; maintained BERT accuracy
4	Johnson et al. (2017) – FAISS	GPU k-selection / similarity search	8.5× faster than prior GPU SOTA	Search speed
5	Guu et al. (2020) – REALM	Retrieval-Augmented Language Model	4–16 percentage-point improvement	Open-domain QA accuracy
6	Olujimi & Ade-Ibijola (2023)	Systematic review of NLP customer-service methods	N/A	Review of 73 studies
7	Gao et al. (2023) – RAG Survey	Naive / Advanced / Modular RAG	N/A	Survey/benchmark review
8	Wang et al. (2024) – Best Practices in RAG	Comparative RAG configurations	No single universal accuracy	Multiple QA/efficiency metrics
9	Es et al. (2024) – RAGAs	Automated RAG evaluation framework	N/A	Evaluation framework
10	Li et al. (2024) – Self-Route	RAG + Long-Context routing	Gemini-1.5-Pro: 46.41 Self-Route vs 37.33 RAG	Average benchmark performance

#	Research Paper	Model / Method	Reported performance	Metric
11	Xu et al. (2024) – KG-RAG	Knowledge Graph + RAG	77.6% improvement in MRR; BLEU +0.32	Retrieval/generation
12	González Juclà et al. (2026) – Retrieval Enhancements	RRF, embedding fusion, LLM reranking	>92% accurate/comprehensive responses when relevant docs are in top 3	Downstream response quality
13	Barrionuevo-Valenzuela et al. (2026) – Agentic-RAG	Multi-agent routing + retrieval validation + generation	Improvement over naïve RAG; no single % reported	Quality/relevance/accuracy

Proposed Model: LLM-Based Intelligent Customer Support Chatbot with Knowledge Retrieval

Stage	Model / Algorithm	Metric	Your Current Result
Product Selection	Product-aware filtering	Product mapping accuracy	~100%*
Query Embedding	all-MiniLM-L6-v2	Recall@5	100.00%
		MRR	99.29%
		Composite embedding/retrieval score	99.72%
Vector Retrieval	FAISS IndexFlatL2	Precision@5	68.40%
		Recall@5	100.00%
		MRR	99.29%
		FAISS composite score	~90.19%
Context Construction	RAG	Context Relevance	100.00%
Response Generation	Google Gemini	Answer Relevance	92.00%
Complete System	Product-Aware RAG	Composite Performance	~93.17%

